STS3215 Register Address Reference

EEPROM Registers (Non-Volatile)

These values are saved even after power off. Torque must be disabled to write.

The **Default** column is what two factory-fresh 12V STS3215 units (firmware 3.10) reported when read back; both units matched on every register. Your unit or firmware may differ, so read the register before relying on a default.

Addr	Name	Size	R/W	Default	Description
0	Firmware Major	1	R	-	Major firmware version number
1	Firmware Minor	1	R	-	Minor firmware version number
3	Servo Major	1	R	-	Servo series major version
4	Servo Minor	1	R	-	Servo series minor version
5	ID	1	R/W	1	Servo ID (0-253), must be unique on bus
6	Baud Rate	1	R/W	0	0=1Mbps, 1=500K, 2=250K, 3=128K, 4=115200, 5=76800, 6=57600, 7=38400
7	Return Delay	1	R/W	0	Response delay time in microseconds
8	Response Level	1	R/W	1	0=No response, 1=Respond to read, 2=Respond to all
9	Min Angle Limit	2	R/W	0	Minimum position limit (0-4095)
11	Max Angle Limit	2	R/W	4095	Maximum position limit (0-4095)
13	Max Temp Limit	1	R/W	70	Maximum temperature limit in °C
14	Max Voltage Limit	1	R/W	140	Maximum voltage limit (value × 0.1V)
15	Min Voltage Limit	1	R/W	40	Minimum voltage limit (value × 0.1V); 40 = 4.0V
16	Max Torque Limit	2	R/W	1000	Maximum torque output (0-1000 = 0-100%)
18	Phase	1	R/W	12	Motor phase/direction setting (factory value; do not change)
19	Unload Condition	1	R/W	44	Conditions that release torque; same bits as Servo Status. 44 = temperature, current, overload
20	LED Alarm	1	R/W	47	Conditions that blink the LED; same bits as Servo Status. 47 = voltage, sensor, temperature, current, overload
21	P Coefficient	1	R/W	32	PID proportional gain
22	D Coefficient	1	R/W	32	PID derivative gain
23	I Coefficient	1	R/W	0	PID integral gain
24	Min Startup Force	2	R/W	16	Minimum torque to start moving (0-1000)

Addr	Name	Size	R/W	Default	Description
26	CW Dead Zone	1	R/W	1	Clockwise dead zone width
27	CCW Dead Zone	1	R/W	1	Counter-clockwise dead zone width
28	Protection Current	2	R/W	310	Overcurrent protection threshold (×6.5mA); 310 ≈ 2.0 A
30	Angular Resolution	1	R/W	1	Position resolution multiplier
31	Position Offset	2	R/W	85	Position offset calibration value (both test units read 85)
33	Mode	1	R/W	0	0=Position, 1=Constant speed, 2=PWM, 3=Step servo
34	Protection Torque	1	R/W	20	Torque during overload protection (%)
35	Protection Time	1	R/W	200	Time before overload protection triggers
36	Overload Torque	1	R/W	80	Torque threshold for overload detection (%)

RAM Registers (Volatile)

These values reset on power cycle. Used for real-time control and status.

Addr	Name	Size	R/W	Description
40	Torque Enable	1	R/W	0=Torque off (servo is limp), 1=Torque on (servo holds position)
41	Acceleration	1	R/W	Goal acceleration setting (0=instant, 1-254=smoother ramp). Note: This is a setting register, not real-time feedback.
42	Goal Position	2	R/W	Target position (0-4095); in motor mode this is speed/direction
44	Goal Time	2	R/W	Time to reach goal position in milliseconds
46	Goal Speed	2	R/W	Speed limit for movement (0=max speed, 1-4095=limited)
48	Torque Limit	2	R/W	Current torque limit (0-1000 = 0-100%)
55	Lock	1	R/W	0=EEPROM unlocked, 1=EEPROM locked (default). While locked, writes to EEPROM addresses still take effect but are not saved across a power cycle
56	Present Position	2	R	Current actual position (0-4095)
58	Present Speed	2	R	Current movement speed
60	Present Load	2	R	Current load/torque (bits 0-9=magnitude, bit 10=direction)
62	Present Voltage	1	R	Current voltage (value × 0.1V)
63	Present Temperature	1	R	Current temperature in °C
64	Async Write Flag	1	R	Status of asynchronous write operation
65	Servo Status	1	R	Error bitmask: bit 0 voltage, 1 sensor, 2 temperature, 3 current, 5 overload
66	Moving	1	R	1=Servo is currently moving, 0=Stationary
69	Present Current	2	R	Current draw (value × 6.5mA)

Quick Reference — Common Operations

Operation	Register(s)	Example Value	Notes
Enable torque	40	1	Must enable before servo responds to goal positions
Disable torque	40	0	Servo goes limp; can move by hand, still read position
Move to position	42	0-4095	2048 = center position
Set speed limit	46	0-4095	0 = maximum speed
Read position	56	-	Returns current position 0-4095
Read load	60	-	Returns load magnitude and direction
Set acceleration	41	0-254	0 = instant, higher = smoother ramp (setting only, no real-time feedback)
Motor (wheel) mode	33	1	`WheelMode()` writes this; angle limits are untouched
Servo (position) mode	33	0	Default; mode reverts to 0 on power cycle unless EEPROM is unlocked (55 = 0) first

Notes

- **Size**: Number of bytes (1 or 2). Use `read1ByteTxRx`/`write1ByteTxRx` for 1-byte, `read2ByteTxRx`/`write2ByteTxRx` for 2-byte registers.
- **Position range**: 0-4095 represents 0-360°. Each unit is approximately 0.088°.
- **Default baud rate**: 1,000,000 bps (1 Mbps).
- **Default servo ID**: 1.
- **Voltage values**: Multiply by 0.1 to get volts (e.g., 74 = 7.4V).
- **Current values**: Multiply by 6.5 to get milliamps.
- **Torque/Load values**: 1000 = 100% of maximum.
- **EEPROM writes**: Disable torque first, then write, then re-enable torque.